

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456776.73691 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.011
Transit depth (Rp/Rs)^2	3.38 +/- 0.42 [%]
Orbital Inclination (inc)	82.74 +/- 1.27
Airmass coefficient 1 (a1)	1.29 +/- 0.017
Airmass coefficient 2 (a2)	-0.1247 +/- 0.0044
Scatter in the residuals of the lightcurve fit is	1.27 %
Best Comparison Star	None
Optimal Aperture	3.65
Optimal Annulus	7.96
Transit Duration (day)	0.0541 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.184 $\pm$ 0.011 vs 0.1594 (deviation 2.24 σ)
Duration fit vs published	0.0541 d vs 0.0512 d (deviation 0.79 σ)
Mid-time O-C	-1.5 min
Depth SNR	16.7
Residual scatter	1.27 %

Light curve

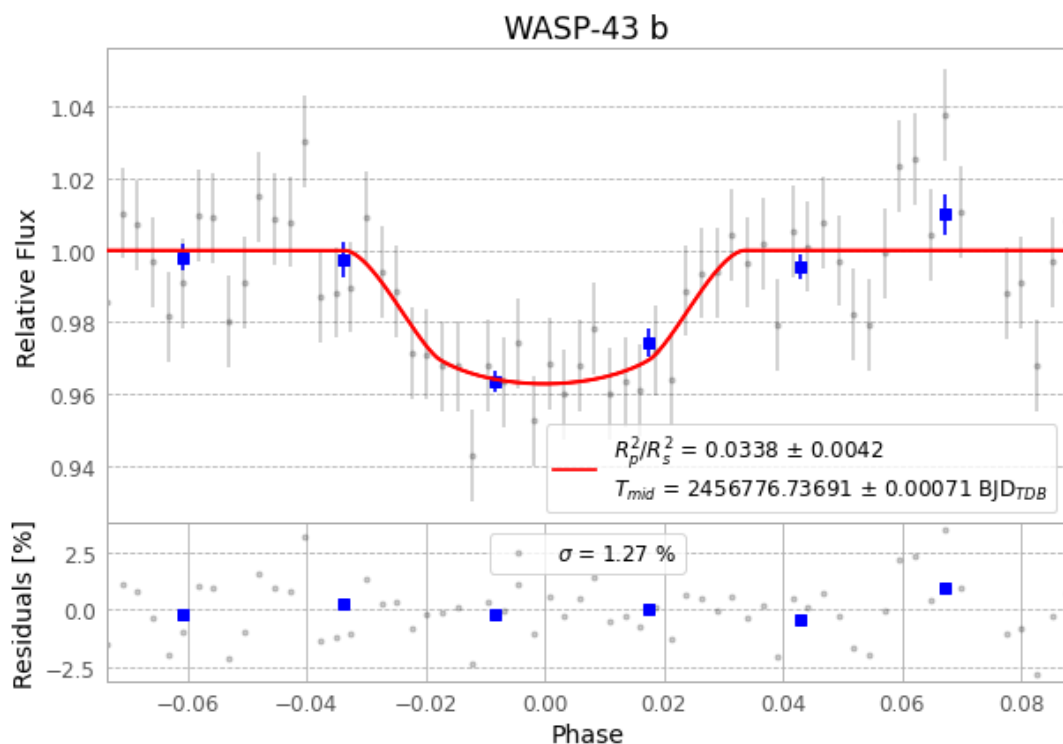


Figure 1 — FinalLightCurve_WASP-43 b_2014-04-28.png (SHA-256 294c81c0b884a3d5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [333, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9102903e70a44c0c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140429040915.FITS → WASP-43140429071812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140429.log (9bbb404ec8c8...), FinalLightCurve_WASP-43 b_2014-04-28.png (294c81c0b884...), FinalLightCurve_WASP-43 b_2014-04-28.csv (8b0a160792d0...)

Compute resources

Wall time 118.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 00:55Z.